Bayesian Data Analysis for Agriculture Science: What, Why, and How

Facilitators:

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Overview

Bayesian data analysis addresses common limitations in agricultural datasets such as small/uneven sample sizes and missing data but is underused in agricultural research. This workshop will introduce Bayesian data analysis, why it is useful, and how to implement it in R using the 'brms' package. You will hear researchers share real-world examples illustrating the use of Bayesian methods in agricultural research and you will implement your own Bayesian data analysis project, with the option of bringing your own analysis-ready dataset. Participants must bring a laptop computer with the necessary components installed ahead of time (instructions provided here).

Pre-workshop preparation:

Please review the background videos available at the link above. Bring a laptop with R, RStudio, and Stan installed and running as well as a research question and dataset (or planned dataset) relevant to your research that might benefit from a Bayesian analytical approach.

Objectives for Day 1: What and Why.

1) Understand what is meant by Bayesian data analysis. Including:
    - Bayes' theorem and its application to linear models.
    - The important differences between Bayesian and frequentist methods.
    - Why modern computers make Bayesian data analysis tractable for complex real-world research applications.
2) Understand why Bayesian data analysis is useful for agricultural research.
    - Kate White: Network-level experimental synthesis with DRIVES.
    - Quadri Popoola: The development of a soil quality index for Northeastern Nigeria
    - Frederi Viens: An example: Bayesian yield prediction in Malawi

Objectives for Day 2: How.

1) Learn a process for building statistical models from conceptual models.

2) Explore tools used to implement a Bayesian data analysis workflow in R and resources for continued self-directed learning.
2) Complete a mini Bayesian data analysis project using a sample agricultural dataset.
3) Plan and start a Bayesian data analysis for your own dataset or research question.

Agenda

Day 1: 10/31/2026, 3:00 PM - 5:00 PM

Time	**Topic/Activity**
3:00 - 4:00 PM	Theory and fundamentals: ● Welcome and workshop overview. ● Why Bayes? ● Basics of Bayesian inference and comparison to frequentist methods ● Bayesian theory demonstration with a simple linear model ● Bayesian computation overview
4:00 - 4:15 PM	Coffee break and computer setup troubleshooting
4:15 - 5:00 PM	Applications: ● Kate White: Network-level experimental synthesis with DRIVES. ● Quadri Popoola: Bayesian Correlation - PCA Approach for Soil quality assessment. ● Frederi Viens: An example: Bayesian yield prediction in Malawi

Day 2: 11/1/2026, 9:00 AM–4:00 PM

9:00 - 10:00 AM	Bayesian tools and workflow: ● Overview of computational tools, workflow, and learning resources ● Setup and installation troubleshooting
10:00 - 10:15 AM	Coffee break
10:15 AM - 12:00 PM	Bayesian data analysis:

	● Overview of the process ● Start mini Bayesian analysis project with sample code
12:00 - 12:45 PM	Lunch break
12:45 - 2:00 PM	Continue mini Bayesian analysis project with sample code
2:00 - 2:15 PM	Coffee break
2:15 - 4:00 PM	Plan/conceptualize a Bayesian data analysis project with your own data